ACS APPLIED MATERIALS & INTERFACES

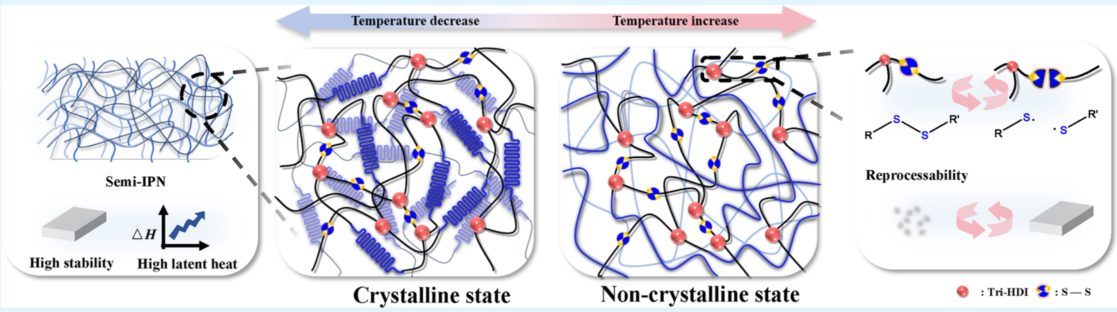
www.acsami.org Research Article

High Latent Heat and Recyclable Phase-Change Materials for Photothermoelectric Conversion

Xiaohui Zeng, Silong Chen, Yansheng Liao, Tianren Liu, Zhimeng Liu, Liang Jiang, Yuan Lei,* Jingxin Lei, and Xiaowei Fu*

Cite This: ACS Appl. Mater. Interfaces 2025, 17, 27155−27166 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information



ABSTRACT: Realizing organic phase-change material networks with excellent recyclability while maintaining high latent heat still faces great challenges due to the difficult trade-off between network composition. Herein, the high latent heat and recyclable phase-change materials (RPCMs) were developed by integrating linear poly(ethylene glycol) (PEG) multimers as phase-change components and the dynamic covalent cross-linking polyurethane network components in a semi-interpenetrating polymer network structure. The latent heat (54.7−145.0 J/g) can be readily tuned by the weight fraction of linear PEG multimers and the molecular weight of PEGs used in RPCMs. The RPCMs have excellent recyclability/reprocessability with activation energy of 81.23−125.84 kJ/mol by introducing dynamic disulfide bonds in the cross-linking network structure components in RPCMs while enabling the adjustable mechanical stress (10.72−14.04 MPa) and strain (7.40−266.22%), high thermal reliability, and high thermal stability. Efficient photothermal RPCMs were realized by introducing polydopamine particles in the RPCM matrix. The thermal management system and the photothermoelectric generator were designed based on the advantages of high latent heat and efficient photothermal ability of RPCMs and further demonstrated the excellent thermal management ability and photothermoelectric properties.

KEYWORDS: *phase-change material, semi-interpenetrating structure network, reprocessability, photothermoelectric conversion, dynamic covalent bond*

1. INTRODUCTION

The development of renewable energy and the highly efficient utilization of existing fossil energy resources have become a global topic due to the emergence of the energy crisis, leading to a strong need for energy storage technologies.[1−3] Latent-heat-based thermal energy storage is one of the most promising energy storage technologies due to its high energy storage density and small temperature fluctuation.[4−6] Organic phase-change materials (PCMs), including paraffin, poly(ethylene glycol) (PEG), etc., have attracted much attention because they have high latent heat, tunable phase-change temperature, tunable mechanical properties, and commercial availability.[7]

PEG is preferred among organic PCMs due to its reactive end groups, which allow for molecular structure design by covalent linkage, and the other intrinsic advantages mentioned above.[8−11] However, traditional PEGs as the phase-change substrate exhibit a transition from the solid state to the liquid state at their phase-change temperature, leading to leakage issues. To address the leakage issue, PEGs usually work as phase-change components for the construction of polymeric PCM networks by the covalent bonding of urethane, an ester linkage.[12−19] For example, Kong et al. synthesized solid−solid PCMs using PEG and polyaryl poly(methylene isocyanate), and they had a maximum latent heat of 112.3 J/g, which is significantly lower than the latent heat (182.9 J/g) of PEG used for the preparation of PCM networks.[20] Fu et al. synthesized PEG-based polyester networks with a maximum latent heat of 120.0 J/g.[21] The introduced cross-linking network structure restricts molecular mobility beyond their

Received: February 13, 2025
Revised: April 16, 2025
Accepted: April 16, 2025
Published: April 28, 2025

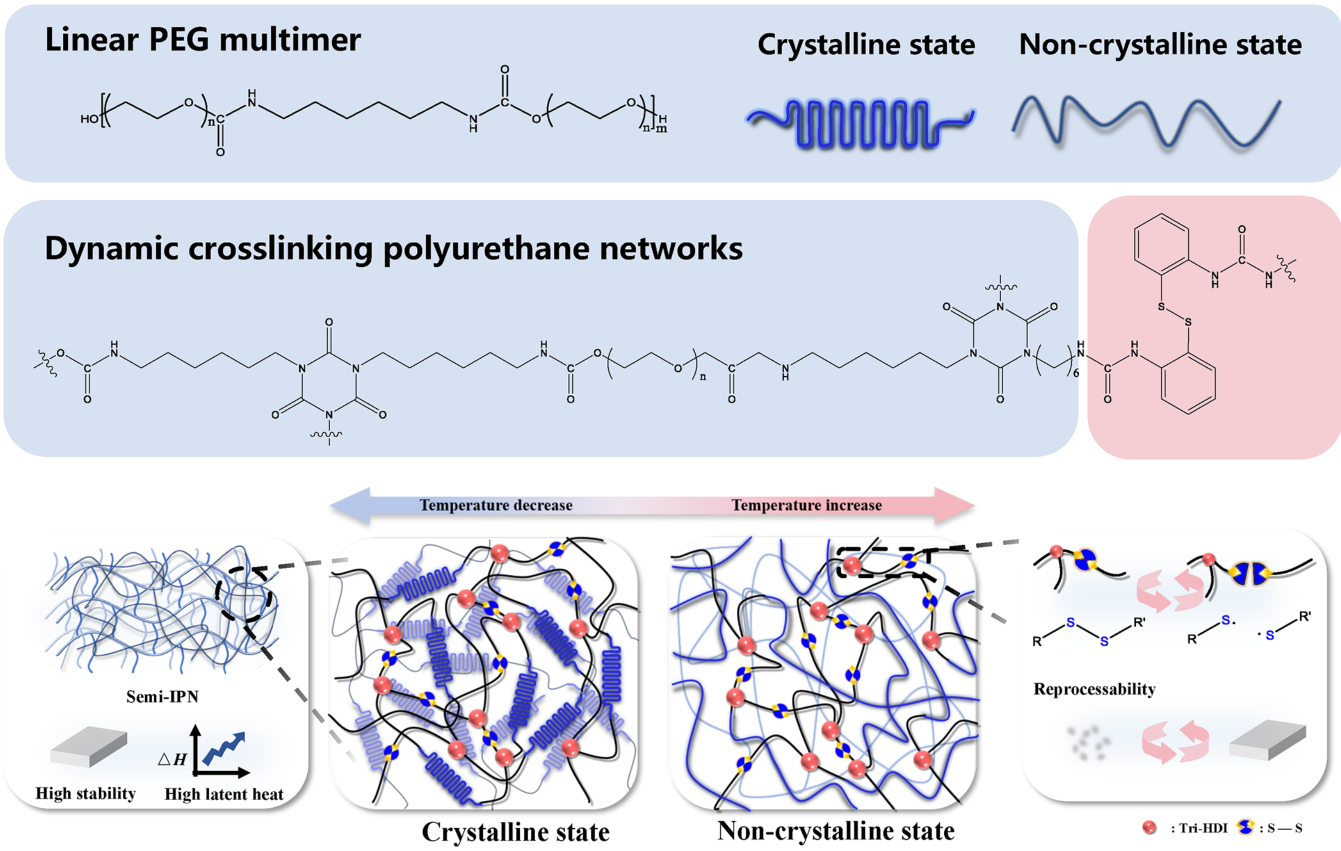


Figure 1. Construction of an RPCM with a dynamic semi-IPN structure. Chemical structures of (a) a linear PEG multimer as the phase-change component and (b) the dynamic cross-linking polyurethane network. (c) Mechanism of the latent-heat-based thermal energy storage/release and recycling/reprocessing of RPCMs.

phase-change temperature, not only successfully solving the leakage issue but also enhancing the mechanical strength and toughness of the material.[22] Meanwhile, the introduced cross-linking network structure reduces the crystallization ability of PEG as a phase-change component, leading to the low latent heat and thus deteriorating the energy storage capacity of PCMs.[23−25] Therefore, introducing a cross-linking network structure can effectively solve the leakage issue of PEGs as traditional solid−liquid PCMs and yet results in the low latent heat of PEG-based PCM networks due to the nature of the cross-linking network structure.

To further address the leakage issue of traditional PCMs while maintaining their high latent heat, PCMs with a semi-interpenetrating network (semi-IPN) structure were developed by our groups and others, where linear PEGs as phase-change components and one cross-linking network work as basic elements for a semi-IPN structure.[26−28] For example, Zhao et al. synthesized semi-IPN-based PCM networks with linear PEG multimers as phase-change components and a cross-linking polyester network, maintaining a nonflowing behavior during the phase-change process while possessing a high latent heat of 141.0 J/g.[29] In the semi-IPN structure, the linear PEG components and cross-linking network components are physically interpenetrated without covalent bonding.[30,31] When the linear PEG phase-change component reaches its melting temperature (T_m), the presence of the cross-linking network structure can reduce the movement of molecular chains and maintain a nonflowing feature of the whole semi-IPN-based PCM network.[29,32] Meanwhile, in the semi-IPN structure, the impact on the crystallization of PEGs as phase-change components is much smaller compared to the direct addition of cross-linking points to the polymer backbone in the polymeric PCM network mentioned above, thus maintaining the original high latent heat of the PEG phase-change components.[33,34] However, semi-IPN-based PCM networks cannot be recycled/reprocessed due to the nature of the permanent covalent cross-linking network structure, restricting their sustainable use. Therefore, introducing a semi-IPN structure can address the leakage issue while maintaining high latent heat compared to conventional polymeric PCM networks, and yet there is still the problem of being unable to recycle/reprocess due to the essence of permanent covalent cross-linking.[35]

Introducing dynamic covalent bonds in PCM networks is the main and important solution for realizing recycling/reprocessing. For example, Huang et al. introduced dynamic oxime−carbamate bonds to PCM networks to achieve reprocessability, but the latent heat was only 88.26 J/g.[36] Kong et al. synthesized PCM networks with dynamic disulfide bonds, enabling reprocessability, and yet the maximum latent heat was only 105.2 J/g.[37] However, PCM networks with introduced dynamic covalent bonds have low latent heat resulting from the nature of the cross-linking network structure, although introducing dynamic covalent bonds can realize recycling and reprocessing due to the essence of dynamic bonds.[38−41] Therefore, to achieve high latent heat and reprocessability while addressing the leakage issue of PCMs, a semi-IPN structure with a dynamic covalent cross-linking

https://doi.org/10.1021/acsami.5c02961
ACS Appl. Mater. Interfaces 2025, 17, 27155−27166

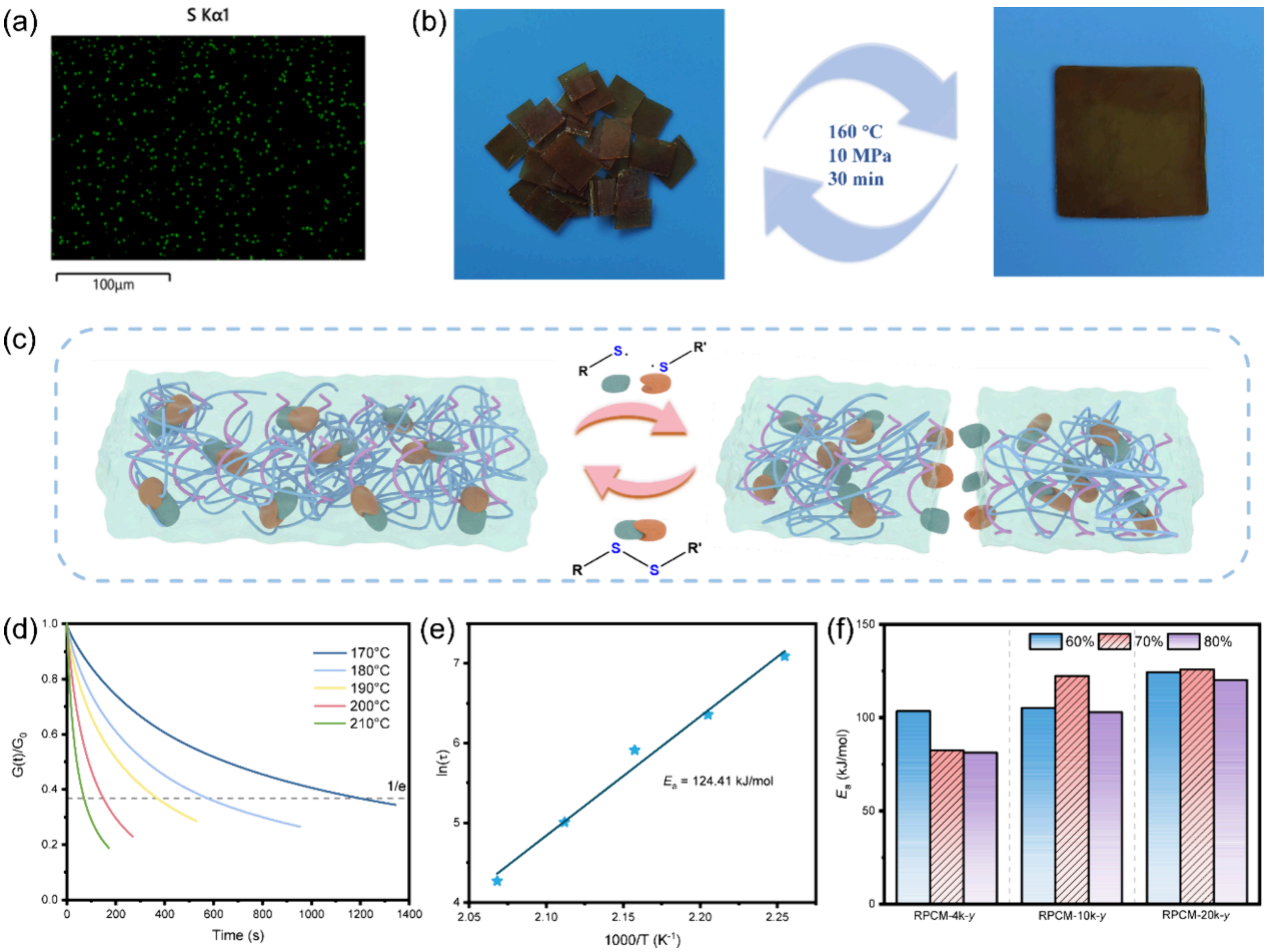


Figure 2. Reprocessability of RPCMs. (a) EDS mapping of S elements in RPCM-10k-70%. (b) Cracked RPCM-4k-60% reformed into whole film samples after reprocessing at 160 °C for 30 min and with a pressure of 10 MPa. (c) Schematic diagram of the reprocessing mechanism of RPCMs. (d) Stress relaxation curves of RPCM-20k-60%. (e) Activation energy of RPCM-20k-60%. (f) Activation energies of RPCM-*x*-*y*.

network will be used to construct the PCMs, enabling the combination of the advantages of semi-IPN-based PCMs and dynamic covalent bonds.

On the basis of the above hypothesis, in this work, we designed and synthesized semi-IPN-based high latent heat and recyclable PCMs (RPCMs) by integrating linear PEG multimers into dynamic covalent cross-linking polyurethane networks with high activity of dynamic disulfide bonds for the first time. The RPCMs had a latent heat of up to 145.0 J/g while enabling recycling/reprocessing with an activation energy of 81.23−125.84 kJ/mol and leak resistance. The RPCMs presented tunable mechanical properties due to the introduction of linear PEG multimers as phase-change components. The enhanced photothermal properties of RPCMs can be realized by adding polydopamine (PDA) nanoparticles.[42,43] Based on the high latent heat and photothermal effect, the thermal management system and photothermoelectric generator were designed.

2. RESULTS AND DISCUSSION

2.1. Synthesis and Reprocessability of RPCMs. Linear hydroxyl-terminated PEG multimers as phase-change components were formed through the addition polymerization of PEG and hexamethylene diisocyanate (HDI; Figure 1a and Scheme S1). The dynamic polyurethane network structure was created by reacting a hexamethylene diisocyanate trimer (Tri-HDI)-terminated PEG prepolymer and a chain extender, 2,2′-diaminodiphenyl disulfide (DADS), in the presence of linear PEG multimers (Figure 1b and Scheme S2). This led to the formation of RPCM-*x*-*y* product with a semi-IPN structure due to the higher reactivity of the isocyanate group with the amino group than with the hydroxyl group, as evidenced by Fourier transform infrared spectroscopy (FTIR; Figures 1c and S1 and Table S1), where x and y represent the number-average molecular weight (M_n) of PEG used for the preparation of linear PEG multimers and the weight fraction of linear PEG multimers in the whole product, respectively.[44]

In RPCMs, the linear PEG multimers as the phase-change components reversibly transform between a crystalline state and a noncrystalline state when the ambient temperature reversibly changes beyond or below its T_m, which is involved in latent-heat-based energy storage and release with a small temperature variation (Figure 1c). In the process, the cross-linking polyurethane network structure effectively prevents leakage of the linear PEG multimers as phase-change components. Additionally, the chain extender, DADS, contains dynamic covalent disulfide bonds that can reform after

https://doi.org/10.1021/acsami.5c02961
ACS Appl. Mater. Interfaces 2025, 17, 27155−27166

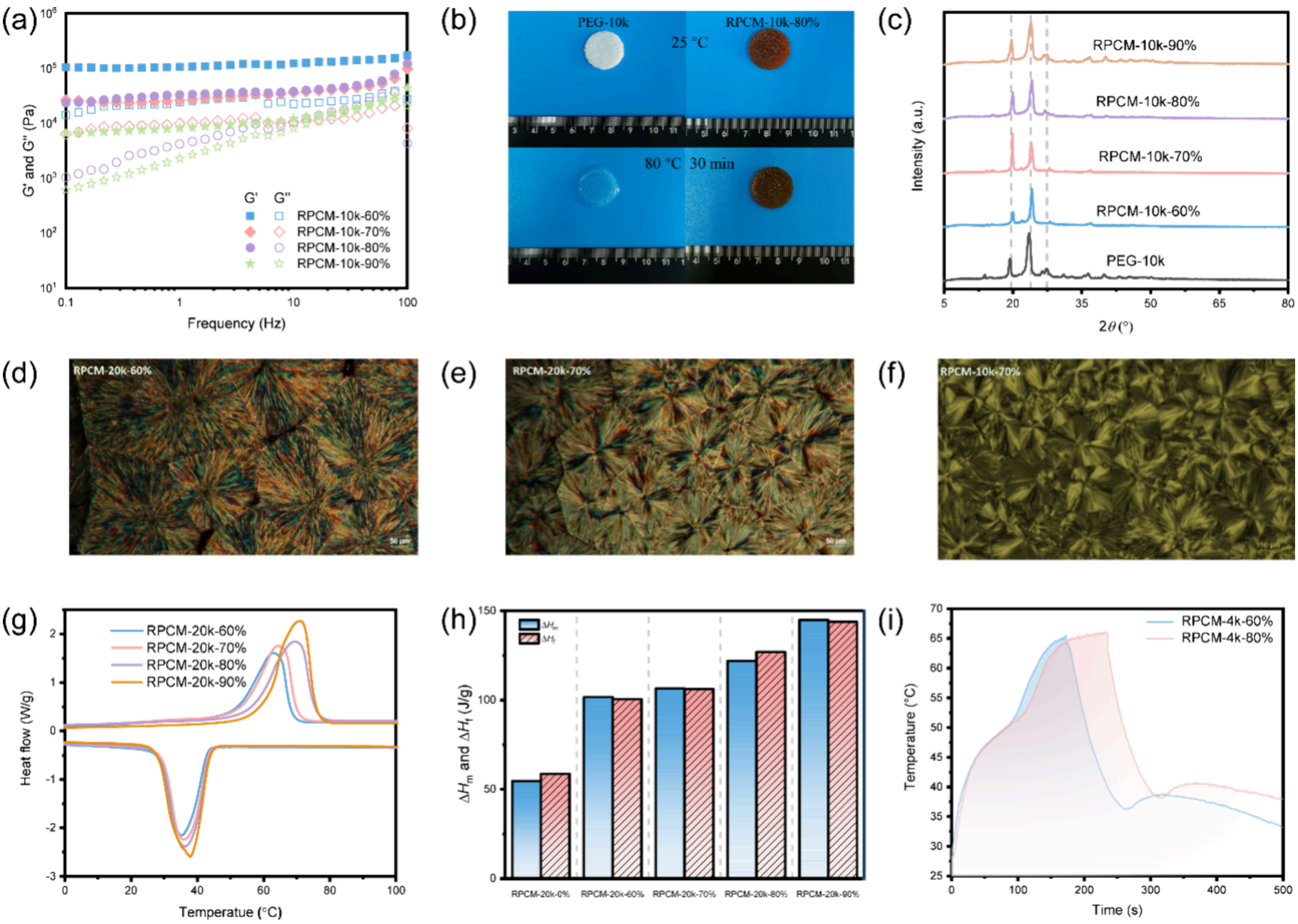


Figure 3. Phase-change properties of RPCMs. (a) Storage (G') and loss (G'') moduli of RPCM-10k-y at a testing temperature of 80 °C. (b) Optical images of PEG-10k and RPCM-10k-80% at 25 and 80 °C for 30 min. (c) XRD curves of RPCM-10k-y and PEG-10k. POM images of (d) RPCM-20k-60%, (e) RPCM-20k-70% and (f) RPCM-10k-70%. (g) DSC curves of RPCM-10k-y. (h) ΔH_m and ΔH_f of RPCM-10k-y (when y = 0, it represents RPCM-10k-0 without linear PEG multimers). (i) Temperature evolution curves of RPCM-4k-60% and RPCM-4k-80% in a heating process at 80 °C and in a cooling process at 25 °C.

breakage, providing the material with recycling and reprocessing properties and making it sustainable.[45−48]

Energy-dispersive X-ray spectroscopy (EDS) determined the presence of the S element of dynamic covalent disulfide bonds derived from the chain extender DADS in RPCMs compared to the disulfide-bond-free control sample, PCM-10k-70%, with 2,2′-(ethane-1,2-diyl)dianiline replacing DADS (Figures 2a and S2, Scheme S3, and Table S2). The dynamic covalent disulfide bonds can enable reprocessing of RPCMs. For example, the cracked RPCM-4k-60% was reprocessed into a whole film sample at 160 °C for 30 min and with a pressure of 10 MPa, proving the reprocessability of RPCMs (Figures 2b and S3). The disulfide-bond-based reprocessing mechanism was proposed: when high temperature and pressure were applied to RPCMs, dynamic covalent disulfide bonds can be cleaved to sulfur radicals and then quickly re-form into new disulfide bonds, leading to dynamic exchangeability and reprocessability of RPCMs (Figure 2c).

The exchange dynamics of RPCMs was further evidenced by stress relaxation behaviors and the corresponding activation energies (Figures 2d−f and S4−S11). The characteristic relaxation time (τ^*) decreased with temperature, indicating that higher temperature induces a higher relaxation rate.[27] For example, the τ^* values of RPCM-20k-60% were 1200.5, 576.5, 369.5, 1149.5, and 71.5 s when the testing temperatures were 170, 180, 190, 200, and 210 °C, respectively. In RPCM-x-y, the activation energy presented a decreasing trend with increasing weight fractions of linear PEG multimers from 60% to 80%, due to the decreasing cross-linking density from 244 mol/m^3 for RPCM-x-60% to 183 mol/m^3 for RPCMs-x-70% to 122 mol/m^3 for RPCMs-x-80%. Also, the activation energy increased with the M_n of PEG used for the preparation of linear PEG multimers in RPCMs. For instance, the activation energy increased from 103.40 kJ/mol of RPCM-4k-60% to 104.35 kJ/mol of RPCM-10k-60% and then to 124.41 kJ/mol of RPCM-20k-60%. This increase is attributed to the entanglement of the molecular chain of the linear PEG multimer, which enhances intermolecular interactions and restricts molecular motion. These results indicate that the reprocessing of RPCMs can be enabled by dynamic covalent disulfide bonds introduced in network components in the semi-IPN structure but cannot be realized in semi-IPN-based PCMs with conventional covalent cross-linking networks.[29,49−51]

2.2. Phase-Change Properties of RPCMs. The cross-linking network structure was used to ensure the shape-stable ability of RPCMs during the whole phase-change process due to osmotic pressure.[52,53] The network structure was confirmed

https://doi.org/10.1021/acsami.5c02961
ACS Appl. Mater. Interfaces 2025, 17, 27155−27166

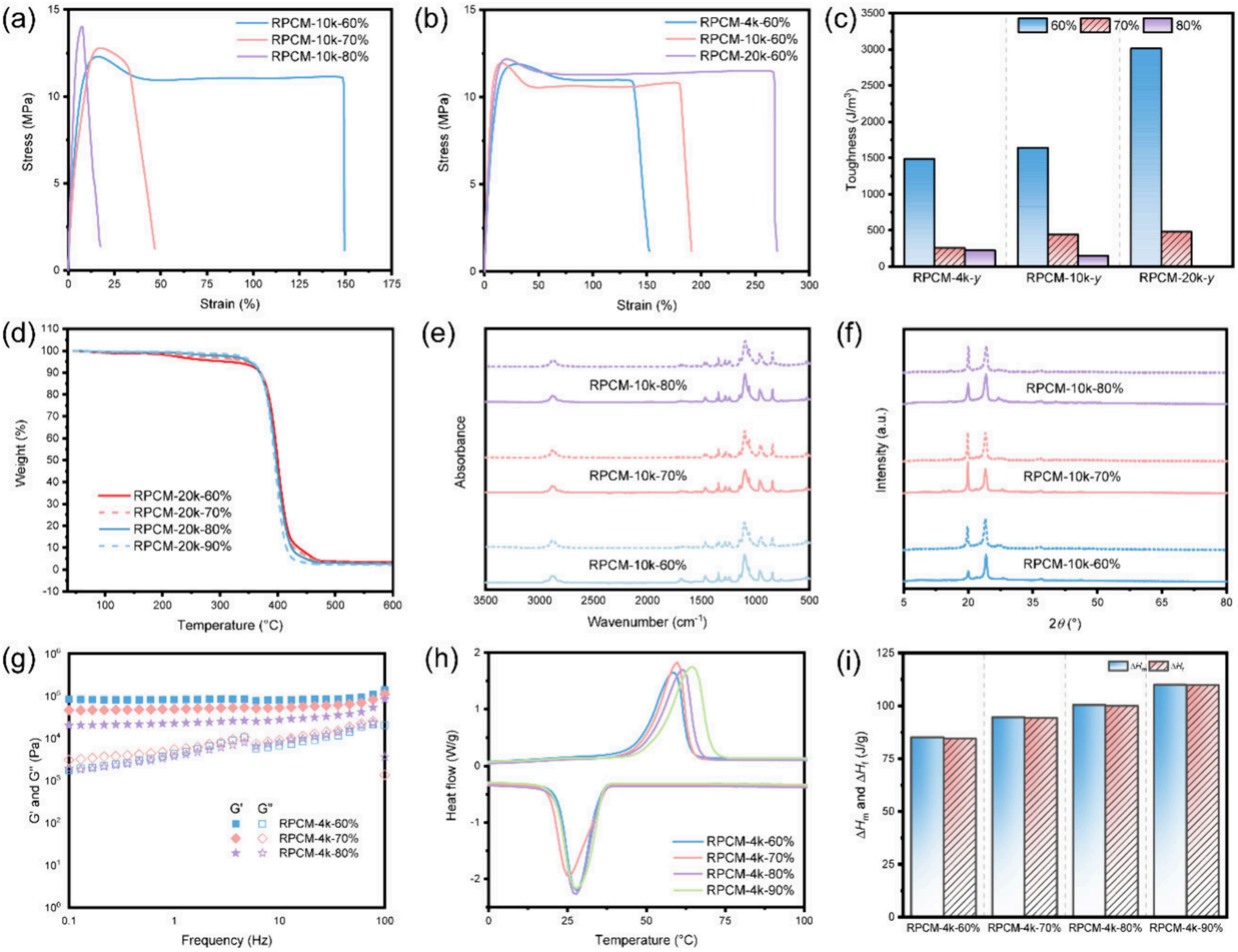


Figure 4. Mechanical properties, thermal stability, and thermal reliability of RPCMs. Stress−strain curves of (a) RPCM-10k-*y* and (b) RPCM-*x*-60%. (c) Toughness of the RPCMs. (d) TGA curves of RPCM-20k-*y*. (e) FTIR spectra and (f) XRD curves of RPCM-10k-*y* before (solid line) and after (dashed line) ATC. (g) Storage (G') and loss (G'') moduli of RPCM-4k-*y* after ATC. (h) DSC curves and (i) corresponding ΔH_m and ΔH_f values of RPCM-4k-*y* after ATC.

through the higher storage modulus (G') than the corresponding loss modulus (G'') at 80 °C beyond T_m (T_m = ∼55 °C; Figures 3a, S12, and S13).[54] When the temperature increased from 25 to 80 °C, the original solid PEG-10k became liquid, indicating the leaking behavior. However, RPCMs remained nonleaking during the phase-change process, indicating that the linear PEG multimers as phase-change components do not leak out of the cross-linking network in RPCMs at 80 °C due to the osmotic pressure (Figures 3b and S14).

The crystalline structures of the linear PEG multimers of RPCMs were confirmed by X-ray diffraction (XRD) analysis at 25 °C, which were the structural basis of the phase-change properties (Figures 3c and S15). RPCMs exhibit distinct diffraction peaks at 2θ = 27.24°, 23.74°, and 19.52°, corresponding to the (210), (112), and (120) crystal planes, which are consistent of those of the original PEGs, indicating that linear PEG multimers as phase-change components retain their crystalline structure.[51] The Maltese cross-extinction phenomenon, caused by the birefringent nature and radial optical symmetry of the spherulite, can be clearly observed in the polarizing optical microscopy (POM) images (Figure 3d−f). In RPCM-*x*-*y*, an increase in *y* leads to an increase in the number of spherulites in the same field of view when *x* is consistent (Figure 3d,e), indicating that the spherulite size decreases, which was not consistent with our previous fatty alcohols-based semi-IPN PCMs.[27] This is because the increase in phase-change components makes the molecular chains more likely to form smaller crystal nuclei, thereby increasing the number of spherulites and reducing the space available for their growth. Conversely, when *y* is consistent and *x* increases, the proportion of molecular chains with regular and ordered structures increases during the crystallization process. This results in the formation of larger spherulites, which occupy more space and reduce the space available for new spherulites, thereby decreasing their number (Figure 3e,f).[55]

The phase-change properties of RPCMs were evidenced by the reversible endothermal and exothermal peaks of differential scanning calorimetry (DSC; Figures 3g and S16). The melting latent heat (ΔH_m) and freezing latent heat (ΔH_f) of RPCMs increased with increasing weight fraction of linear PEG multimers as the phase-change components independent of the M_n of PEGs used for the linear PEG multimers in RPCMs. Taking RPCM-20k-*y* as an example, ΔH_m elevated from 54.7 to 101.7 J/g with increasing *y* from 0% to 60%, and then to

145.0 J/g with further increasing y to 90% in RPCM; especially, RPCM-20k-0% presented a ΔH_m of 54.7 J/g due to the nature of the PEG-based cross-linking polyurethane networks (Figure 3h and Tables S4 and S5). For RPCM-x-y, ΔH_m promoted to 2.03, 2.49, and 2.65 times those of ΔH_m of a neat dynamic polyurethane network structure, named as RPCM-x-0%, without linear PEG multimers with increasing y from 0% to 90% when x was 4k, 10k, and 20k, respectively. The results indicate that linear PEG multimers work as the main-phase change components and a dynamic polyurethane network structure served in the preparation of semi-IPN RPCMs. ΔH_m of RPCM-x-90% augmented from 111.3 to 136.1 to 145.0 J/g when x changed from 4k to 10k to 20k. This phenomenon is due to the increased M_n of PEG in the linear PEG multimers because the phase-change component facilitates the formation of a more regular crystalline structure, and thus more latent heat is required along with a reversible transformation between the crystalline and amorphous states during the heating and freezing processes. The highest ΔH_m of 145.0 J/g for RPCM-20k-90% in this report is much higher than ΔH_m of conventional covalent cross-linking PCM networks[20,56,57] and semi-IPN-based PCMs with pure PEGs as the phase-change components[29] due to the nature of the semi-IPN-based structure of RPCMs and the linear PEG multimers working as phase-change components in RPCMs; meanwhile, RPCMs have reprocessability compared to conventional covalent cross-linking networks and semi-IPN-based networks with phase-change properties due to the introduction of dynamic covalent disulfide bonds in RPCMs (Table S6).[58] The ΔH_m change exhibited by RPCM-4k-60% before and after reprocessing is within 5%, indicating that its phase-change properties remain basically unchanged before and after reprocessing (Figure S17 and Table S7).

The temperature evolution curves of RPCMs were used to evaluate the small temperature variation during the phase-change process (Figure 3i). The RPCMs exhibited a rise in temperature once the RPCM was transferred from 25 °C of the hot plate onto 80 °C of the hot plate. Especially, the rate of temperature increase slowed down between 34 and 100 s, totaling 66 s, demonstrating the latent-heat-based energy storage derived from the melting of phase-change components of RPCM-4k-60%. As such, during the whole cooling process, a slowing temperature decrease was observed along with a slight warming subprocess once the temperature dropped below the freezing temperature (T_f = ∼36 °C). This is due to the fact that crystallization of RPCMs produced heat, leading to a slight rise of temperature, followed by a sustained decline. The latent-heat-based energy storage and release processes of RPCMs-4k-y prolonged with increasing y from 60% to 80% due to the increasing latent heat of RPCMs. This demonstrates that the resulting RPCMs can be thermally managed on a macroscopic level.

2.3. Mechanical Properties, Thermal Stability, and Thermal Reliability of RPCMs. The mechanical properties of RPCMs were evaluated by the tensile stretching testing (Figures 4a,b and S18 and Table S8). The tensile stress−strain curves of RPCM-10k-y changed from ductile fracture behavior to brittle fracture behavior with increasing the weight fraction of linear PEG multimer from 60% to 70% to 80%. For instance, the tensile strength reached 12.30 MPa for RPCM-10k-60%, 12.80 MPa for RPCM-10k-70%, and 14.04 MPa for RPCM-10k-80%, showing an increase in tensile strength due to the higher yield strength. Meanwhile, the elongation at break reached 148.51% for RPCM-10k-60%, 31.28% for RPCM-10k-70%, and 7.40% for RPCM-10k-80%, indicating an obvious decreasing trend. This is due to the following: (1) the increasing linear PEG multimers as phase-change components result in a decrease in dynamic cross-linking polyurethane components, producing the low cross-linking density; (2) the cross-linking polyurethane components have more high strength urea bonds that form more hydrogen bonds than conventional carbamate bonds, significantly improving the tensile strength and elongation at break.

In addition, with M_n of the used PEGs increasing from 4 to 20 kDa for RPCM-x-60%, the elongation at break increases from 134.02% to 266.22%, while the tensile strength remained nearly constant (Figure 4b). Especially, the tensile toughness of RPCM-x-60% enhanced from 14.88 MJ/m^3 for x = 4k to 16.38 MJ/m^3 for x = 10k to 30.16 MJ/m^3 for x = 20k compared to RPCM-x-70% and RPCM-x-80% (Figure 4c). This is because the high M_n of PEG used leads to an increase in the molecular length of linear PEG multimers as phase-change components and thus produces high interactions between the linear PEG multimers and the covalent cross-linking polyurethane network, leading to high tensile strain and toughness.

The thermal stability of RPCMs was evaluated by thermogravimetric analysis (TGA; Figures 4d and S19 and Table S9). All RPCMs showed an initial degradation of less than 5% when the temperature increased to 308.37 °C due to the delicate disulfide bonds and their derivatives. When the temperature rose to 355.39 °C, all RPCMs exhibited significant degradation due to the chain-scission degradation of the whole RPCMs. The degradation temperature of all RPCMs was much higher than the corresponding phase-change temperature and the reprocessing temperature, indicating that the RPCMs have good thermal stability.

Accelerating thermal cycling (ATC) testing was employed in a high/low-temperature chamber with alternating heating and cooling cycles for 100 times from 20 to 80 °C.[59−61] The thermal reliability of RPCMs was verified by comparing the chemical structure (Figures 4e and S20), crystalline structure (Figures 4f and S21), network conditions (Figures 4g and S22), and phase-change properties (Figure 4h,i) before and after ATC. The characteristic absorption peaks of RPCMs have the same position and comparable intensity before and after ATC, indicating that the chemical structure of RPCMs did not change before and after ATC. The diffraction peaks appeared near 2θ = 19.80°, 23.74°, and 27.04°, corresponding to the (210), (112), and (120) crystal planes of PEG, respectively, demonstrating that the crystal structure of RPCMs remains unchanged before and after ATC. The higher G' than G'' verified that the cross-linking structure of the RPCMs did not change before and after ATC (Figure 4g). The DSC curves of RPCM-4k-y after ATC showed the reversible melting and freezing processes, that is, the reversible phase-change properties, and the ΔH_m change was less than 4.12% before and after ATC (Figure 4h,i and Table S10). Therefore, these results demonstrate that RPCMs have high thermal reliability.

2.4. Imparting Photothermal Properties to RPCMs. The PDA molecule contains a large number of conjugated double-bond structures as well as the phenolic hydroxyl and amine groups, enabling a strong photothermal effect and strong adhering ability between PDA and the polymer matrix, respectively, so PDA was introduced in RPCMs to impart a photothermal performance.[62−64] Fe(III)-chelated PDA nano-

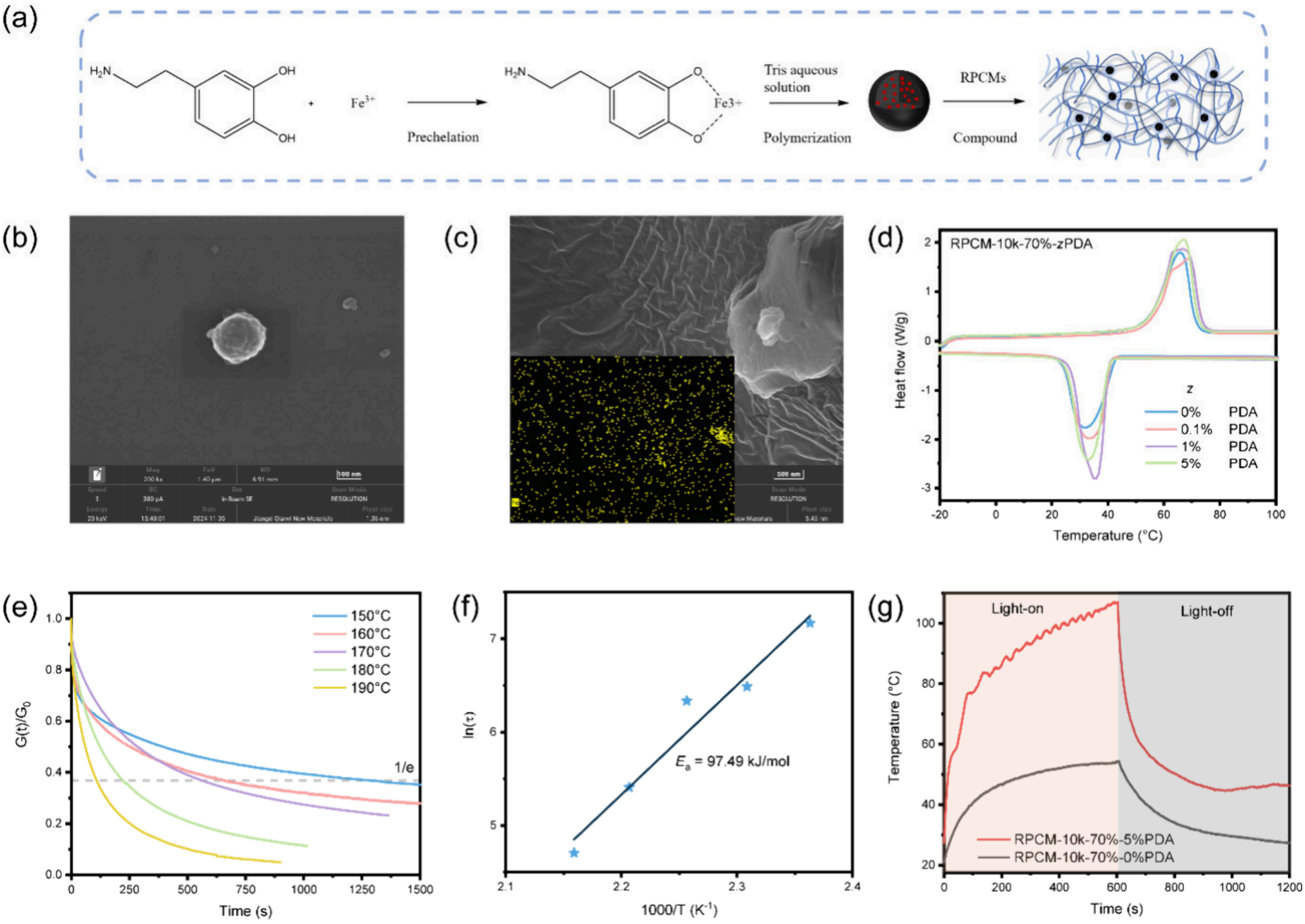


Figure 5. Synthesis and properties of PDA: (a) synthesis of PDA; (b) SEM image of PDA particles. Properties of RPCMs with a photothermal effect: (c) SEM image and Fe elemental mapping of RPCM-10k-70%−1% PDA; (d) DSC curves of RPCM-10k-70%-*z*PDA; (e) stress relaxation curves and (f) corresponding Arrhenius fitting curve of RPCM-10k-70%−1%PDA; (g) temperature evolution curves of RPCM-10k-70%−5%PDA with light on/off.

particles were synthesized using a prechelation−polymerization method, resulting in PDA particle sizes ranging from 200 to 500 nm (Figures 5a,b and S23) and PDAs having an initial degradation temperature of 200.24 °C with high mass residue of 25.4% (Figure S24). The photothermal RPCM-10k-70%-*z*PDA (where *z* represents the weight fraction of PDA relative to the polymer matrix) were prepared by introducing PDA particles in a RPCM matrix, as evidenced by the SEM morphology, Fe element mapping, and FTIR spectra (Figures 5c, S25, and S26). The crystalline structure of RPCM-10k-70%-*z*PDA was analyzed by XRD, revealing diffraction peaks at 2θ = 19.64°, 23.72°, and 26.60°, corresponding to the (210), (112), and (120) crystalline planes, respectively (Figure S27). The POM image of RPCM-10k-70%−5%PDA confirmed the Maltese cross-extinction phenomenon (Figure S28). This observation confirms that the introduction of PDA nanoparticles does not change the crystalline structure and spherulite morphology.

The phase-change properties of RPCM-10k-70%-*z*PDA were analyzed by DSC (Figure 5d and Table S11). When the mass fraction of PDA increased from 0% to 5%, ΔH_m of the material increased from 104.4 to 114.9 to 124.3 to 125.6 J/g. This increase is attributed to the added PDA acting as a nucleating agent, promoting the formation of a more regular crystal structure in the phase-change component.[65−68] The dynamic exchangeability of RPCM-10k-70%−1%PDA was analyzed using the stress relaxation curves (Figure 5e). The characteristic relaxation time decreased with the temperature due to the thermal activation of molecular chain movement. In particular, the activation energy of RPCM-10k-70%−1%PDA was found to be 97.49 kJ/mol, which is 20.31% lower compared to RPCM-10K-70% without PDA (E_a = 122.33 kJ/mol; Figures 5f and S8). This reduction is due to the introduced PDA nanoparticles having a large amount of heteroatom such as oxygen and nitrogen atoms on the surface, activating the dynamic disulfide bonds and then decreasing the activation energy.[69] Additionally, the dynamic disulfide-bond-based reprocessing of RPCM-10k-70%−1%PDA was demonstrated by hot-pressing the sheared samples into a whole film sample (Figure S29).

The photothermal properties of RPCMs with PDA added were evaluated by the photothermal temperature evolution curves recording the surface temperature when using the sunlight simulator (light intensity of 500 mW/cm^2) to irradiate RPCM-10k-70%−5%PDA (Figures 5g and S30). RPCM-10k-70%−5%PDA reached 107.11 °C after 600 s of light irradiation, which was 52.66 °C higher compared to that of the control sample RPCM-10k-70%. During the photothermal process, a small temperature plateau of ∼23 s was observed due to the phase-change operation, and yet this was a very

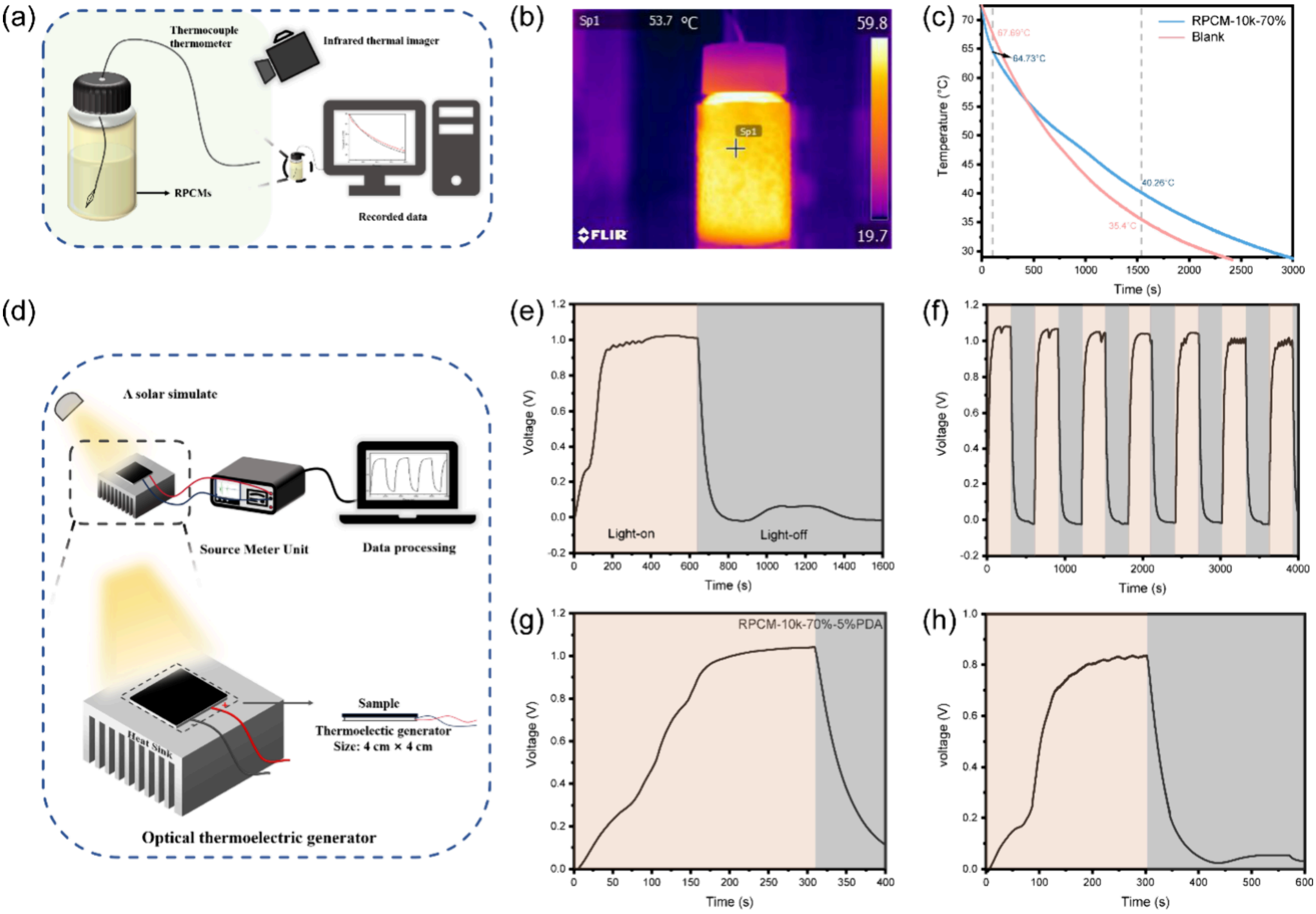


Figure 6. Thermal management systems and photothermoelectric applications. (a) Schematic diagram of the thermal management system. (b) Infrared thermogram of the thermal management system during operation. (c) Change of the liquid temperature with time under the thermal management system versus room temperature. (d) Schematic diagram of the photothermoelectric device. (e, f) Voltage response of a photothermoelectric device with RPCM-10k-70%−5%PDA used by turning on/off the light source at (e) low and (f) high change frequency. (g) Capacitor charging and discharging processes after the photothermoelectric device is connected to an external capacitor. (h) Load voltage when the photothermoelectric device drive motor is operating.

short time span for the temperature plateau compared to the direct-thermal temperature evolution curves because of the fast surface temperature rise effect from photothermal operation. When irradiation was stopped, the subsequent temperature change showed an overall decreasing trend. However, there was a small increase in temperature, followed by a decrease, which was due to the exothermic nature of the phase-change component crystallization. To investigate the photothermal self-healing properties, the surface of RPCM-10k-70%−5% PDA was scratched with a wound width of ∼78 μm (Figure S31). Remarkably, after 1 drop of *N*,*N*-dimethylformamide (DMF) was applied to the scratches, followed by 30 min of light irradiation, the scratches were visibly healed. The photothermal self-healing mechanism was analyzed as follows: upon light irradiation, the temperature at the scratching of the sample increases and activates the dynamic exchange in the sample, leading to photothermal healing. These results indicate that introducing PDA particles can endow RPCMs with efficient photothermal properties, further enabling photothermal healing.

2.5. Thermal Management Systems and Photothermoelectric Applications. Based on the high latent heat of RPCMs and enhanced photothermal properties by introducing PDA in RPCMs, a thermal management system and a photothermal power generation device were designed to expand their applications. To simulate the thermostatic effect of RPCMs in the thermal management system, RPCM-10k-70% was used as the wrapping material to wrap a glass bottle with a volume of 20 mL (Figure 6a), and the thickness of the wrapping layer was 2.88 mm. The glass bottle was filled with hot water with a temperature beyond 60 °C, and the temperature change of water inside the bottle with time was monitored (Figure 6b,c). Its comparison with the temperature change of water inside the glass bottle without wrapped RPCMs (Figure 6c) shows that there is an intersection point between the two temperature curves at about $t = 420$ s. When $t < 420$ s, the water temperature inside the glass bottle with RPCMs wrapped decreases faster, which is the PCM absorbing the thermal energy of the water inside the bottle rapidly and storing it; meanwhile, the water temperature decreases from 72.50 to 56.45 °C. During this period, the maximum temperature difference between the water temperature in the glass bottle with and without RPCMs wrapped is 3.23 °C, and at this time, the water temperatures of the two are 64.73 and 67.69 °C. As the temperature continues to fall to 56.34 °C, the water temperature in the bottle with RPCMs wrapped begin to slowly decline compared to the water temperature in the bottle without RPCMs wrapped. At this time, the maximum

https://doi.org/10.1021/acsami.5c02961
ACS Appl. Mater. Interfaces 2025, 17, 27155−27166

temperature difference is 4.86 °C, which is 40.26 and 35.4 °C for the bottles with and without RPCMs wrapped, respectively. The temperature at this point (T = 40.26 °C) is just below the freezing temperature (T_f = 42.80 °C) at which RPCMs-10k-70% begin to crystallize (Figure 3g). This is due to the large amount of heat released from the RPCMs, which slows the rate of decrease of the water temperature in the bottle. When both water temperatures drop to 30 °C, the times consumed for the bottle with and without RPCM wrapped are 2762 and 2243 s, respectively. This indicates that the RCPMs have a good thermal management effect.

A photothermoelectric generator was prepared by combining photothermal RPCM-10k-70%−5%PDA and Seebeck-effect-based thermoelectric panels (size: 4 cm × 4 cm), which utilizes temperature differences for thermoelectric conversion (Figure 6d).[70,71] The voltage variation with light on/off was achieved by controlling the irradiation time (Figure 6e). With an increased irradiation time, the voltage gradually increased, reaching up to 1.02 V. During the voltage increase, there was a zone of slow voltage growth lasting about 22.43 s, corresponding to the phase change plateau zone when the surface temperature changed under light conditions. Additionally, there was a slight increase in the voltage after it decreased to the lowest point when the light stopped. This corresponds to the temperature change curve of the RPCMs, where the thermal energy released from crystallization of the RPCMs is converted to electrical energy. This situation did not occur with RPCM-10k-70% as a substrate, and there were no RPCMs due to the absence of an efficient photothermal effect (Figure S32).

The voltage changed periodically when light irradiation was turned on and off periodically, indicating high reproducibility (Figure 6f). The electrical energy was stored in the capacitor when connecting the photothermoelectric generator to the capacitor and was released through electric circuit construction (Figures 6g and S33). Also, the electrical energy generated by the photothermoelectric device was converted into mechanical energy by directly connecting a small fan to the photothermoelectric generator (Figure 6h).

3. CONCLUSION

In summary, we successfully developed a new class of RPCMs with high latent heat and reprocessability based on a semi-IPN-based structure composed of linear PEG multimers as the phase change components and dynamic polyurethane network components. The RPCMs can be reprocessed by hot-press with activation energies of 81.23−125.84 kJ/mol due to the dynamic disulfide bonds used to construct the covalent cross-linking polyurethane network components. The RPCMs have a tunable latent heat of 85.4−145.0 J/g depending on the weight fraction of linear PEG multimers and the M_n of PEG used in the semi-IPN-based structure. The highest latent heat of 145.0 J/g for the RPCMs resulted from the semi-IPN-based structure and the linear PEG multimers as phase-change components. The RPCMs present adjustable tensile strength (10.72−14.04 MPa) and elongation at break (7.40−266.22%) depending on the weight fraction and composite of linear PEG multimers in RPCMs. The RPCMs have high thermal reliability and high thermal stability, with the main degradation temperature beyond 300 °C. The enhanced photothermal properties can be obtained by introducing PDA particles to the RPCMs, enabling efficient photothermal healing of scratches, relying on the essence of dynamic exchangeability and photothermal ability of photothermal RPCMs. The thermal management system was designed based on the high latent heat of RPCMs, realizing thermal management such as slowing down the temperature reduction of the target substance. Further, a photothermoelectric generator was completed by combining the photothermal RPCMs and Seebeck-effect-based thermoelectric panels, enabling multiple conversions from solar energy to thermal energy and then to electrical energy. The electrical energy generated by a photothermoelectric generator can be stored in a capacitor, following the subsequent release when needed, and was directly used to drive a motor realizing a further transformation to mechanical energy. The RPCMs provide a robust design platform for high latent heat, and RPCMs and photothermal RPCMs with the introduction of PDA provide a feasible solution to the imbalance of solar irradiation in time and space in the future.

4. EXPERIMENTAL SECTION

Using RPCM-4k-60% as an example, 11.51 g (2.88 mmol, 1.01 equiv) of melted PEG-4k and 0.48 g (2.85 mmol, 1 equiv) of HDI were added and reacted at 60 °C for 1 h and then further reacted at 100 °C for 4 h to obtain the linear hydroxy-terminated PEG multimer, as Liquid A. Separately, 5.82 g (1.46 mmol, 1 equiv) of melted PEG-4k and 2.18 g (3.63 mmol, 2.49 equiv) of Tri-HDI were added to 23.32 g of dried DMF and reacted at 60 °C for 1 h, then at 80 °C for 4 h, and finally at 100 °C for 2 h to obtain an isocyanate-terminated prepolymer, as Liquid B. After that, Liquid A was added to Liquid B at 60 °C, and then DADS was added. The mixture was cast into a Teflon mold and then cured at 80 °C for 5 days. The sample was obtained after the DMF was removed under vacuum.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.5c02961.

Detailed synthetic process of RPCMs and related supporting materials, mechanical properties, phase deformation energy and reprocessability of RPCMs, morphology characterization of PDA, photothermoelectric conversion and reprocessing properties of PRCM-10k-70%-zPDAs, and so on (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Xiaowei Fu − *State Key Laboratory of Polymer Materials Engineering, Polymer Research Institute, Sichuan University, Chengdu 610065, China*; orcid.org/0000-0001-8179-265X; Email: xwfu@scu.edu.cn

Yuan Lei − *State Key Laboratory of Polymer Materials Engineering, Polymer Research Institute, Sichuan University, Chengdu 610065, China*; orcid.org/0009-0000-1331-3100; Email: yuanlei2023@scu.edu.cn

Authors

Xiaohui Zeng − *State Key Laboratory of Polymer Materials Engineering, Polymer Research Institute, Sichuan University, Chengdu 610065, China*

Silong Chen − *State Key Laboratory of Polymer Materials Engineering, Polymer Research Institute, Sichuan University, Chengdu 610065, China*

Yansheng Liao − *State Key Laboratory of Polymer Materials Engineering, Polymer Research Institute, Sichuan University, Chengdu 610065, China*

https://doi.org/10.1021/acsami.5c02961
ACS Appl. Mater. Interfaces 2025, 17, 27155−27166

Tianren Liu − *State Key Laboratory of Polymer Materials Engineering, Polymer Research Institute, Sichuan University, Chengdu 610065, China*
Zhimeng Liu − *School of Chemical Engineering, Sichuan University, Chengdu 610065, China*
Liang Jiang − *State Key Laboratory of Polymer Materials Engineering, Polymer Research Institute, Sichuan University, Chengdu 610065, China;* orcid.org/0000-0003-1305-1926
Jingxin Lei − *State Key Laboratory of Polymer Materials Engineering, Polymer Research Institute, Sichuan University, Chengdu 610065, China*

Complete contact information is available at:
https://pubs.acs.org/10.1021/acsami.5c02961

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

X.F. thanks the Natural Science Foundation of China (Grant 22401205) and Natural Science Foundation of Sichuan Province (Grant 2024NSFSC1029). J.L., X.F., and L.J. thank the Sichuan Province Advanced Building Materials Production Education Integration Innovation Demonstration Platform. L.J. thanks the Natural Science Foundation of Sichuan Province (Grant 2023NSFSC0984). Z.L. thanks the Sichuan Science and Technology Program (Grant 2023NSFSC1131). Yu.L. thanks the China Postdoctoral Science Foundation (Grant 2024M752207) and State Key Laboratory of Polymer Materials Engineering (Grant sklpme2024-2-06).

■ REFERENCES

(1) Aftab, W.; Huang, X.; Wu, W.; Liang, Z.; Mahmood, A.; Zou, R. Nanoconfined phase change materials for thermal energy applications. *Energy Environ. Sci.* **2018**, *11*, 1392−1424.
(2) Aftab, W.; Usman, A.; Shi, J.; Yuan, K.; Qin, M.; Zou, R. Phase change material-integrated latent heat storage systems for sustainable energy solutions. *Energy Environ. Sci.* **2021**, *14*, 4268−4291.
(3) Chen, X.; Cheng, P.; Tang, Z.; Xu, X.; Gao, H.; Wang, G. Carbon-Based Composite Phase Change Materials for Thermal Energy Storage, Transfer, and Conversion. *Adv. Sci.* **2021**, *8*, No. 2001274.
(4) Wang, G.; Tang, Z.; Gao, Y.; Liu, P.; Li, Y.; Li, A.; Chen, X. Phase Change Thermal Storage Materials for Interdisciplinary Applications. *Chem. Rev.* **2023**, *123*, 6953−7024.
(5) Freeman, T. B.; Foster, K. E. O.; Troxler, C. J.; Irvin, C. W.; Aday, A.; Boetcher, S. K. S.; Mahvi, A.; Smith, M. K.; Odukomaiya, A. Advanced Materials and Additive Manufacturing for Phase Change Thermal Energy Storage and Management: A Review. *Adv. Energy Mater.* **2023**, *13*, No. 2204208.
(6) Xie, B.; Du, S.; Wang, R.; Kou, X.; Jiang, J.; Li, C. Heat pump integrated with latent heat energy storage. *Energy Environ. Sci.* **2024**, *17*, 6943−6973.
(7) Tao, J.; Luan, J.; Liu, Y.; Qu, D.; Yan, Z.; Ke, X. Technology development and application prospects of organic-based phase change materials: An overview. *Renew. Sust. Energy Rev.* **2022**, *159*, No. 112175.
(8) Sundararajan, S.; Samui, A. B.; Kulkarni, P. S. Versatility of polyethylene glycol (PEG) in designing solid−solid phase change materials (PCMs) for thermal management and their application to innovative technologies. *J. Mater. Chem. A* **2017**, *5*, 18379−18396.
(9) Wang, C.; Geng, X.; Chen, J.; Wang, H.; Wei, Z.; Huang, B.; Liu, W.; Wu, X.; Hu, L.; Su, G.; Lei, J.; Liu, Z.; He, X. Multiple H-Bonding Cross-Linked Supramolecular Solid−Solid Phase Change Materials for Thermal Energy Storage and Management. *Adv. Mater.* **2024**, *36*, No. 2309723.
(10) Pang, Y.; Sun, J.; Zhang, W.; Lai, C.; Liu, Y.; Guo, H.; Zhang, D. Green, recyclable and high latent heat form-stable phase change composites supported by cellulose nanofibers for thermal energy management. *Int. J. Biol. Macromol.* **2024**, *264*, 130633.
(11) Pang, Y.; Chen, J.; Sun, J.; Zhang, D.; Li, Y.; Hua, Y.; Liu, Y.; Guo, H.; Lai, C.; Song, Y. A flexible phase change composite encapsulated in cellulose nanofiber/ polyacrylamide layered porous structure for thermal energy storage and management. *Industrial Crops Products* **2024**, *222*, 119920.
(12) Usman, A.; Xiong, F.; Aftab, W.; Qin, M.; Zou, R. Emerging Solid-to-Solid Phase-Change Materials for Thermal-Energy Harvesting, Storage, and Utilization. *Adv. Mater.* **2022**, *34*, No. 2202457.
(13) Huang, L.; Yang, Y.; Yuan, D.; Cai, X. Solid-solid phase-change materials with excellent mechanical property and solid state plasticity based on dynamic urethane bonds for Thermal Energy Storage. *J. Energy Storage* **2021**, *36*, No. 102343.
(14) Huan, Y.; Gu, M.; Ni, Y.; Xue, H.; Zhu, H.; Zhu, Y.; Guo, Q.; Fan, D.; Zhou, X.; Liu, J.; Cao, Y.; Lu, Y.; Yan, C.; Qian, T. Highly flexible, healable and degradable polyurethane phase change materials with exceptional mechanical properties for thermal regulation. *Chem. Eng. J.* **2023**, *468*, No. 143742.
(15) Huang, Y.; Luo, W.; Chen, W.; Hu, X.; Zhu, G.; Ma, Y.; Jiang, X.; Li, Q. Self-healing, adaptive and shape memory polymer-based thermal interface phase change materials via boron ester cross-linking. *Chem. Eng. J.* **2024**, *496*, No. 153789.
(16) Zhu, M.; Cojocaru Mirédin, O.; Mio, A. M.; Keutgen, J.; KüPers, M.; Yu, Y.; Cho, J. y.; Dronskowski, R.; Wuttig, M. Unique Bond Breaking in Crystalline Phase Change Materials and the Quest for Metavalent Bonding. *Adv. Mater.* **2018**, *30*, No. 1706735.
(17) Usman, A.; Qin, M.; Xiong, F.; Aftab, W.; Shen, Z.; Bashir, A.; Han, H.; Han, S.; Zou, R. MXene-Integrated Solid-Solid Phase Change Composites for Accelerating Solar-Thermal Energy Storage and Electric Conversion. *Small Methods* **2024**, *8*, No. e2301458.
(18) Li, L.; Peng, J.; Wang, L.; Lai, J.; Zhao, C.; Xiang, D.; Li, H.; Yan, G.; Li, Z.; Wu, Y. Highly flexible GO−polyurethane solid−solid phase change composite materials for efficient photothermal conversion and thermal energy storage. *J. Mater. Chem. A* **2025**, *13*, 3073−3083.
(19) Jing, Y.; Zhao, Z.; Cao, X.; Sun, Q.; Yuan, Y.; Li, T. Ultraflexible, cost-effective and scalable polymer-based phase change composites via chemical cross-linking for wearable thermal management. *Nat. Commun.* **2023**, *14*, 8060.
(20) Kong, W.; Fu, X.; Liu, Z.; Zhou, C.; Lei, J. A facile synthesis of solid-solid phase change material for thermal energy storage. *Appl. Therm. Eng.* **2017**, *117*, 622−628.
(21) Fu, X.; Jiang, L.; Wang, Y.; Xiao, Y.; Wu, B.; Kong, W.; Lei, J. Reprocessable, biodegradable polyester-based solid-solid phase change materials networks from dynamic ionic crosslinking with high latent heat capability. *J. Clean. Prod.* **2021**, *297*, No. 126630.
(22) Deng, J.; Bai, R.; Zhao, J.; Liu, G.; Zhang, Z.; You, W.; Yu, W.; Yan, X. Insights into the Correlation of Cross-linking Modes with Mechanical Properties for Dynamic Polymeric Networks. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202309058.
(23) Kong, W.; Lei, Y.; Jiang, Y.; Lei, J. Preparation and thermal performance of polyurethane/PEG as novel form-stable phase change materials for thermal energy storage. *J. Therm. Anal. Calorim.* **2017**, *130*, 1011−1019.
(24) Sundararajan, S.; Samui, A. B.; Kulkarni, P. S. Crosslinked polymer networks of poly(ethylene glycol) (PEG) and hydroxyl terminated poly(dimethyl siloxane) (HTPDMS) as polymeric phase change material for thermal energy storage. *Sol. Energy* **2019**, *181*, 187−194.
(25) Chen, C.; Chen, J.; Jia, Y.; Topham, P. D.; Wang, L. Binary shape-stabilized phase change materials based on poly(ethylene glycol)/polyurethane composite with dual-phase transition. *J. Mater. Sci.* **2018**, *53*, 16539−16556.

(26) Zeng, X.; Liu, X.; Zhu, H.; Zhu, J.; Lan, J.; Yu, Y.; Lee, Y. S.; Yang, X. Advanced Crosslinked Solid Polymer Electrolytes: Molecular Architecture, Strategies, and Future Perspectives. *Adv. Energy Mater.* **2024**, *14*, No. 2402671.

(27) Wei, Z.; Liao, Y.; Liu, T.; Yuan, A.; Wu, X.; Jiang, L.; Lei, J.; Fu, X. Design of Sustainable Self-Healing Phase Change Materials by Dynamic Semi-Interpenetrating Network Structure. *Adv. Funct. Mater.* **2024**, *34*, No. 2312019.

(28) Lu, Y.; Yue, Y.; Ding, Q.; Mei, C.; Xu, X.; Jiang, S.; He, S.; Wu, Q.; Xiao, H.; Han, J. Environment-tolerant ionic hydrogel−elastomer hybrids with robust interfaces, high transparence, and biocompatibility for a mechanical−thermal multimode sensor. *InfoMat* **2023**, *5*, No. e12409.

(29) Zhao, Y.; Liu, T.; Wei, Z.; Zhao, S.; Lei, J.; Fu, X. Towards high performance semi-interpenetrating phase change materials networks via linear polyethylene glycol-based multimerization effect. *Chem. Eng. J.* **2022**, *446*, No. 136982.

(30) Dai, Z.; Yang, X.; Wu, F.; Wang, L.; Xiang, K.; Li, P.; Lv, Q.; Tang, J.; Dohlman, A.; Dai, L.; Shen, X.; You, L. Living fabrication of functional semi-interpenetrating polymeric materials. *Nat. Commun.* **2021**, *12*, 3422.

(31) Guo, H.; Sanson, N.; Marcellan, A.; Hourdet, D. Thermoresponsive Toughening in LCST-Type Hydrogels: Comparison between Semi-Interpenetrated and Grafted Networks. *Macromolecules* **2016**, *49*, 9568−9577.

(32) Fu, X.; Yuan, A.; Lei, Y.; Wei, Z.; Xu, H.; Lei, J. Thermosetting semi-interpenetrating phase change materials with heat-triggered network self-dissociation. *Sol. Energy Mater. Sol. Cells* **2021**, *225*, No. 111031.

(33) Qian, P.; Li, L.; Wang, H.; Sheng, J.; Zhou, Y.; Shi, H. SPEEK-based composite proton exchange membrane regulated by local semi-interpenetrating network structure for vanadium flow battery. *J. Membr. Sci.* **2022**, *662*, No. 120973.

(34) Zhang, Y.; Xiu, J.; Tang, B.; Lu, R.; Zhang, S. Novel semi-interpenetrating network structural phase change composites with high phase change enthalpy. *AIChE J.* **2018**, *64*, 688−696.

(35) Lei, Y.; Wei, Z.; Wang, Y.; Lei, J.; Fu, X. Dynamic ionic crosslinking, graft polyethylene glycol-based polymeric phase change materials networks with ultrahigh latent heat efficiency and recycling ability. *Polymer* **2023**, *280*, No. 126069.

(36) Huang, R.; Yang, A.; Liu, Z.; Liu, F.; Fang, T.; He, F.; Li, Y.; Jiang, Z.; Lan, H.; Zhang, K.; Yang, W. Poly(ethylene glycol)-based polyurethanes based on dynamic oxime-urethane bonds for sustainable thermal energy storage. *Colloid. Surface. A* **2024**, *698*, No. 134544.

(37) Kong, W.; Yang, Y.; Yuan, A.; Jiang, L.; Fu, X.; Wang, Y.; Xu, H.; Liu, Z.; Lei, J. Processable and recyclable crosslinking solid-solid phase change materials based on dynamic disulfide covalent adaptable networks for thermal energy storage. *Energy* **2021**, *232*, No. 121070.

(38) Hammer, L.; Van Zee, N. J.; Nicolaÿ, R. Dually Crosslinked Polymer Networks Incorporating Dynamic Covalent Bonds. *Polymers* **2021**, *13*, 396.

(39) Yang, A. s.; Huang, R.; Jiang, Z.; Li, Y.; He, F.; Zhang, Q.; Zhou, Y.; Wang, P.; He, G.; Yang, W. Recyclable Solid−Solid Phase Change Materials with Excellent Reprocessing Properties Based on Dynamic Disulfide Bonds. *ACS Appl. Polym. Mater.* **2023**, *5*, 1499−1508.

(40) Xu, Y.; Zhou, S.; Wu, Z.; Yang, X.; Li, N.; Qin, Z.; Jiao, T. Room-temperature self-healing and recyclable polyurethane elastomers with high strength and superior robustness based on dynamic double-crosslinked structure. *Chem. Eng. J.* **2023**, *466*, No. 143179.

(41) Cho, D. H.; Cho, K. G.; An, S.; Kim, M. S.; Oh, H. W.; Yeo, J.; Yoo, W. C.; Hong, K.; Kim, M.; Lee, K. H. Self-healable, stretchable, and nonvolatile solid polymer electrolytes for sustainable energy storage and sensing applications. *Energy Storage Mater.* **2022**, *45*, 323−331.

(42) Chai, Z.; Fang, M.; Min, X. Composite phase-change materials for photo-thermal conversion and energy storage: A review. *Nano Energy* **2024**, *124*, No. 109437.

(43) Cui, X.; Ruan, Q.; Zhuo, X.; Xia, X.; Hu, J.; Fu, R.; Li, Y.; Wang, J.; Xu, H. Photothermal Nanomaterials: A Powerful Light-to-Heat Converter. *Chem. Rev.* **2023**, *123*, 6891−6952.

(44) Wu, T.; Wu, D.; Deng, Y.; Luo, D.; Wu, F.; Dai, X.; Lu, J.; Sun, S. Three-dimensional network-based composite phase change materials: Construction, structure, performance and applications. *Renew. Sust. Energy Rev.* **2024**, *199*, No. 114480.

(45) Takahashi, A.; Goseki, R.; Otsuka, H. Thermally Adjustable Dynamic Disulfide Linkages Mediated by Highly Air-Stable 2,2,6,6-Tetramethylpiperidine-1-sulfanyl (TEMPS) Radicals. *Angew. Chem., Int. Ed.* **2017**, *56*, 2016−2021.

(46) Sölle, B.; Schmallegger, M.; Schlögl, S.; Rossegger, E. Wavelength-Dependent Dynamic Behavior in Thiol−Ene Networks Based on Disulfide Exchange. *J. Am. Chem. Soc.* **2024**, *146*, 34152−34157.

(47) Wojtecki, R. J.; Meador, M. A.; Rowan, S. J. Using the dynamic bond to access macroscopically responsive structurally dynamic polymers. *Nat. Mater.* **2011**, *10*, 14−27.

(48) Tsuruoka, A.; Takahashi, A.; Aoki, D.; Otsuka, H. Fusion of Different Crosslinked Polymers Based on Dynamic Disulfide Exchange. *Angew. Chem., Int. Ed.* **2020**, *59*, 4294−4298.

(49) Wang, M.; Jiang, J.; Liang, S.; Sui, C.; Wu, S. Functional Semi-Interpenetrating Polymer Networks. *Macromol. Rapid Commun.* **2024**, *45*, No. 2400539.

(50) Alizadeh, N.; Broughton, R. M.; Auad, M. L. Graft Semi-Interpenetrating Polymer Network Phase Change Materials for Thermal Energy Storage. *ACS Appl. Polym. Mater.* **2021**, *3*, 1785−1794.

(51) Zhao, Y.; Liu, T.; Wei, Z.; Yuan, A.; Chen, Y.; Jiang, L.; Lei, J.; Fu, X. Polymeric phase change material networks based on multi-telechelic polyethylene glycol-derived multimer structures for thermal energy storage. *Chem. Eng. J.* **2023**, *462*, No. 142164.

(52) Zeng, L.; Fu, Y.; Cui, H.; Zhao, Y.; Liu, Y.; Lin, X.; Chao, T.; Guo, H. Strong, Smart, and Slippery Organo-gels by Network Glassification. *Adv. Funct. Mater.* **2024**, *34*, 2407397.

(53) Fernández-Rico, C.; Schreiber, S.; Oudich, H.; Lorenz, C.; Sicher, A.; Sai, T.; Bauernfeind, V.; Heyden, S.; Carrara, P.; Lorenzis, L. D.; Style, R. W.; Dufresne, E. R. Elastic microphase separation produces robust bicontinuous materials. *Nat. Mater.* **2024**, *23*, 124−130.

(54) Liu, T.; Zhao, Y.; Lei, Y.; Zhao, S.; Jiang, L.; Wang, J.; Fu, X.; Lei, J. Catalyst-free, reprocessable, intrinsic photothermal phase change materials networks based on conjugated oxime structure. *Chem. Eng. J.* **2022**, *450*, No. 138144.

(55) Wang, W.; Hou, G.; Zheng, Z.; Wang, L.; Liu, J.; Wu, Y.; Zhang, L.; Lyulin, A. V. Designing polymer nanocomposites with a semi-interpenetrating or interpenetrating network structure: toward enhanced mechanical properties. *Phys. Chem. Chem. Phys.* **2017**, *19*, 15808−15820.

(56) Du, X.; Jin, L.; Deng, S.; Zhou, M.; Du, Z.; Cheng, X.; Wang, H. Recyclable, Self-Healing, and Flame-Retardant Solid−Solid Phase Change Materials Based on Thermally Reversible Cross-Links for Sustainable Thermal Energy Storage. *ACS Appl. Mater. Interfaces* **2021**, *13*, 42991−43001.

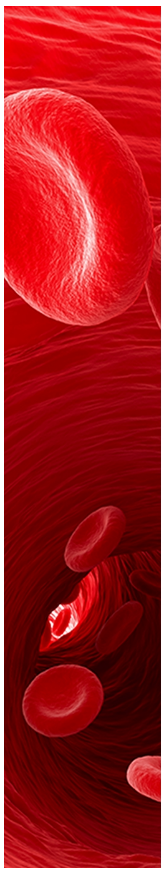
(57) Tian, C.; Ning, J.; Yang, Y.; Zeng, F.; Huang, L.; Liu, Q.; Lv, J.; Zhao, F.; Kong, W.; Cai, X. Super tough and stable solid−solid phase change material based on π-π stacking. *Chem. Eng. J.* **2022**, *429*, No. 132447.

(58) Hao, X. y.; Yu, B.; Li, L.; Ju, H.; Tian, M.; Cao, P. f. Semi-Interpenetrating Polyurethane Network with Fatigue Elimination and Upcycled Mechanical Performance. *Macromolecules* **2024**, *57*, 5063−5072.

(59) Ding, J.; Chen, S.; Zhao, Y.; Wu, X.; Zhao, S.; Yuan, A.; Lei, J.; Fu, X. Engineering Polymeric Phase-Change Sub-Microcapsules with a High Encapsulation Fraction by Facile, Versatile Suspension Polymerization. *ACS Sustain. Chem. Eng.* **2023**, *11*, 14000−14009.

(60) Ding, J.; Liu, T.; Zhao, Y.; Wu, X.; Zhao, S.; Jiang, L.; Lei, J.; Fu, X. Robust, Versatile Access to Quasi-Monodispersed Phase

Change Sub-Microcapsules via Ab Initio Emulsion Polymerization. *ACS Sustain. Chem. Eng.* **2023**, *11*, 1482−1491.
(61) Zhao, S.; Yuan, A.; Chen, X.; Lei, Y.; Fu, X.; Lei, J.; Jiang, L. Single-actuated and fully recyclable phase change materials enabled multiple thermochromism toward information storage and encryption. *Chem. Eng. J.* **2024**, *481*, 148698.
(62) Du, X.; Qiu, J.; Deng, S.; Du, Z.; Cheng, X.; Wang, H. Flame-retardant and solid-solid phase change composites based on dopamine-decorated BP nanosheets/Polyurethane for efficient solar-to-thermal energy storage. *Renew. Energy* **2021**, *164*, 1−10.
(63) Xu, G.; Zhang, H.; Xu, F.; Sun, L.; Lin, X.; Ma, L.; Peng, H.; Wang, Y.; Li, B.; Yan, E. Poly-dopamine coated-cellulose/chitosan hybrid carbon aerogel composite phase change materials with efficient photothermal conversion and storage. *J. Energy Storage* **2024**, *101*, No. 113978.
(64) Xu, N.; Hu, A.; Pu, X.; Li, J.; Wang, X.; Wang, J.; Huang, Z.; Liao, X.; Yin, G. Fe(III)-Chelated Polydopamine Nanoparticles for Synergistic Tumor Therapies of Enhanced Photothermal Ablation and Antitumor Immune Activation. *ACS Appl. Mater. Interfaces* **2022**, *14*, 15894−15910.
(65) Beaupere, N.; Soupremanien, U.; Zalewski, L. Nucleation triggering methods in supercooled phase change materials (PCM), a review. *Thermochim. Acta* **2018**, *670*, 184−201.
(66) Liao, Y.; Wei, Z.; Chen, S.; Zeng, X.; Chen, X.; Lei, Y.; Jiang, L.; Liu, Z.; Lei, J.; Fu, X. Enabling Supramolecular Phase Change Materials with High Latent Heat Ability and Recyclability. *Ind. Eng. Chem. Res.* **2024**, *63*, 14190−14198.
(67) Liu, T.; Zhao, Y.; Xu, H.; Lei, Y.; Yuan, A.; Chen, Y.; Lei, J.; Fu, X. Intrinsic fluorescent phase change materials-based polymer networks: Tuning fluorescence emission intensity and phase change properties for thermal energy storage. *Polymer* **2023**, *268*, No. 125726.
(68) Geng, X.; Wang, C.; Chen, J.; Wang, H.; Liu, W.; Hu, L.; Lei, J.; Liu, Z.; He, X. Phase Change Nanocapsules Enabling Dual-Mode Thermal Management for Fast-Charging Lithium-Ion Batteries. *ACS Nano* **2024**, *18*, 11300−11310.
(69) Zmerli, I.; Michel, J. P.; Makky, A. Bioinspired polydopamine nanoparticles: synthesis, nanomechanical properties, and efficient PEGylation strategy. *J. Mater. Chem. B* **2020**, *8*, 4489−4504.
(70) Chen, S.; Ding, J.; Zhao, Y.; Yuan, A.; Liao, Y.; Chen, X.; Lei, Y.; Jiang, L.; Lei, J.; Fu, X. Design of Robust and Recyclable Covalent Elastomeric Networks through Dynamic Enamine−Urea Bonds. *Macromolecules* **2024**, *57*, 8813−8825.
(71) Wu, X.; Yuan, Y.; Zhao, S.; Lei, Y.; Fu, X.; Lei, J.; Jiang, L. The Synergistic Effects between Liquid Crystal and Crystalline Phase on Photo-Responsive Elastomers toward Quick Photo-Responsive Performance. *Macromol. Rapid Commun.* **2023**, *44*, 2300354.



https://doi.org/10.1021/acsami.5c02961
ACS Appl. Mater. Interfaces 2025, 17, 27155−27166